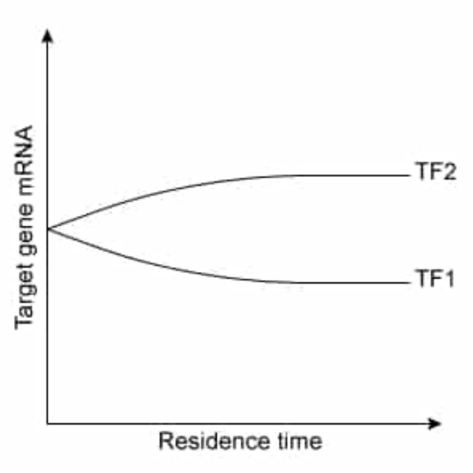
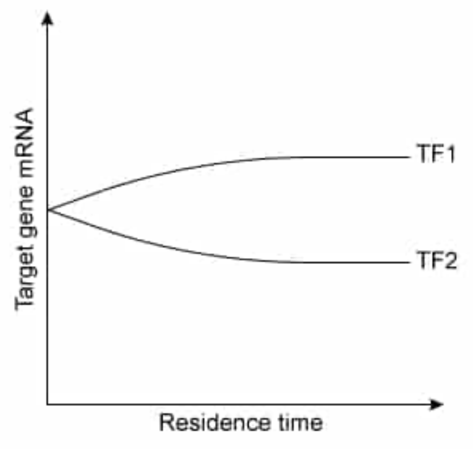
Assume that the effect of transcription factors (TFs) is greater for longer than for shorter periods of physical contact (residence time) with DNA. Which of the following figures shows the effect of residence time on target gene transcription for two TFs: TF1, an activator, and TF2, a repressor?

A.



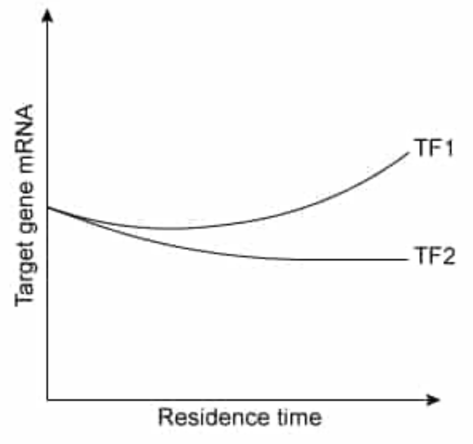
[6%]

✔ B.



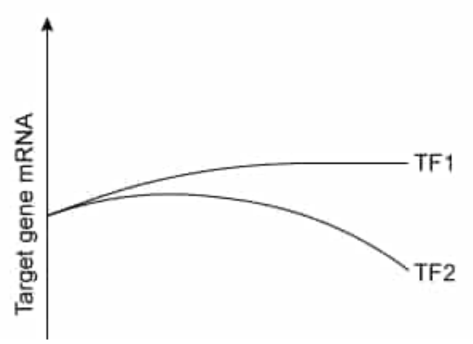
[43%]

C.



[30%]

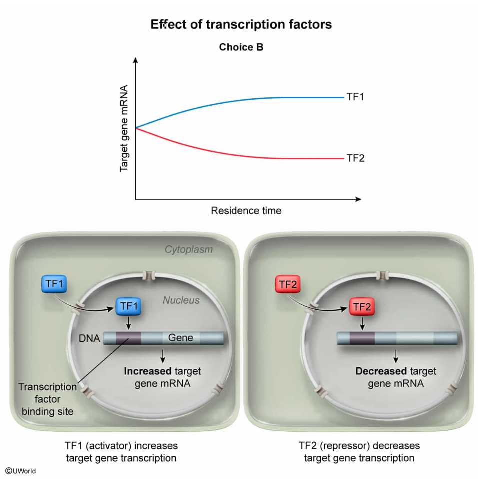
D.



[19%]

Correct
44% answered correctly

Explanation:



Transcription factors (TFs) are proteins that influence transcription by binding to specific DNA sequences in the promoter region for a target gene (ie, a gene regulated by the TF). TFs bind these specific DNA sequences with much greater affinity than other stretches of DNA and such binding can facilitate transcription, in which case the TF is called an activator. In other cases, TF binding can repress transcription, in which case the TF is referred to as a repressor.

The question describes a scenario in which the period of physical contact (residence time) between a TF and DNA influences the effect of that TF on transcription. Specifically, the effect of TFs is greater for longer than shorter periods of residence time with DNA. Therefore, target gene mRNA levels should not be equal for all values of residence time. As **TF1** is defined in the question as an **activator**, **target gene mRNA levels** should **increase** in response to TF1 binding to DNA. Likewise, binding of the **repressor TF2** to DNA should **reduce target gene mRNA levels**. Only the figure in Choice B correctly depicts this scenario.

(Choice A) As an activator, binding of TF1 to a promoter should contribute to increased (*not* decreased)

production of target gene mRNA. As a repressor, TF2 should contribute to decreased (*not* increased) production of target gene mRNA when bound to DNA.

(Choices C and D) These choices depict *both* activation and repression by the *same* transcription factor, but no information is given to suggest such a response. Typically, activators should be considered to promote transcription, and repressors should be considered to repress transcription.

Educational objective:
Transcription factors bind specific promoter DNA sequences to increase (activators) or decrease (repressors) transcription of target genes.

Time spent:0 sec
Subject:
Biology

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System:
Molecular Biology